

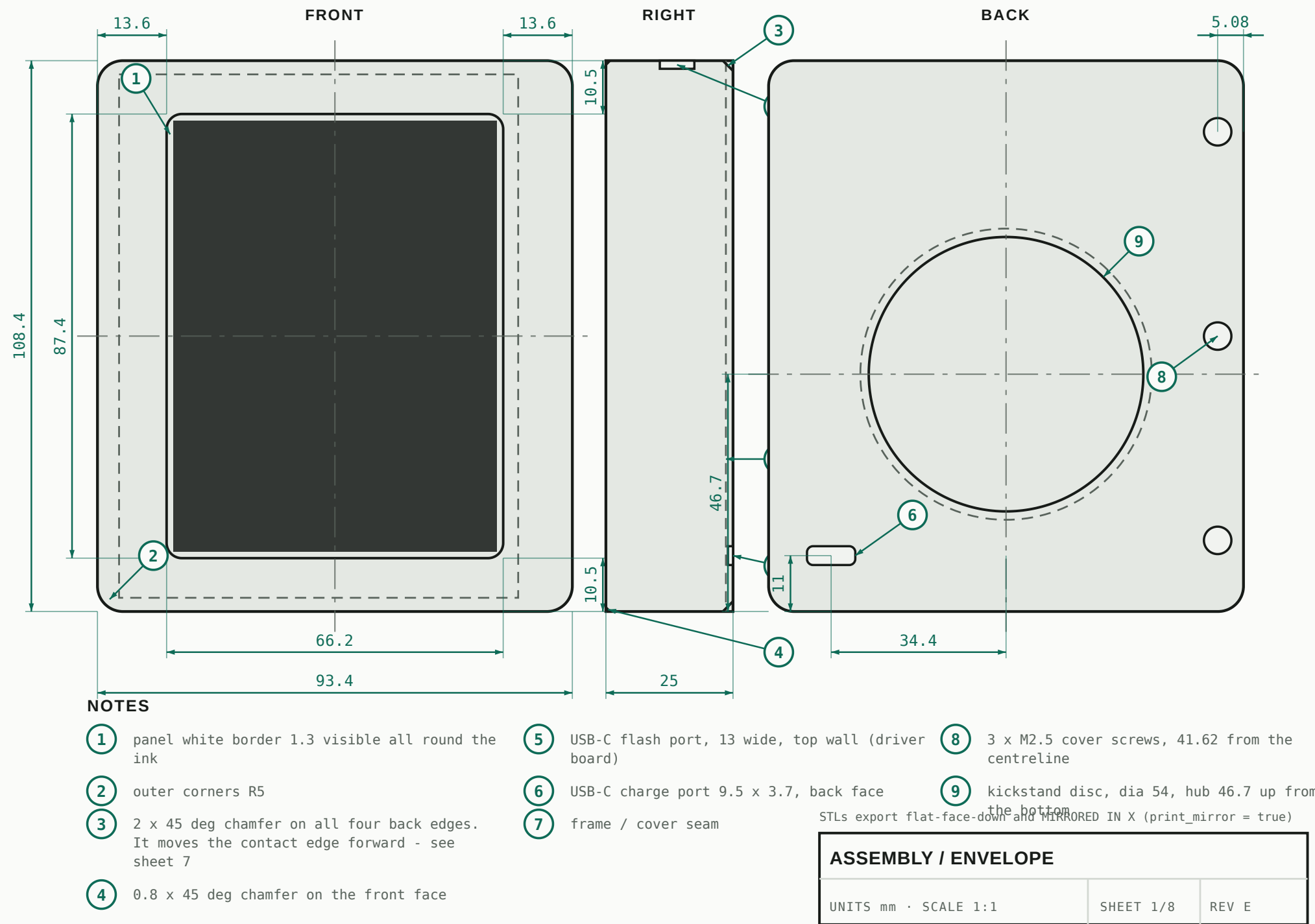
4.2" e-Paper desk frame

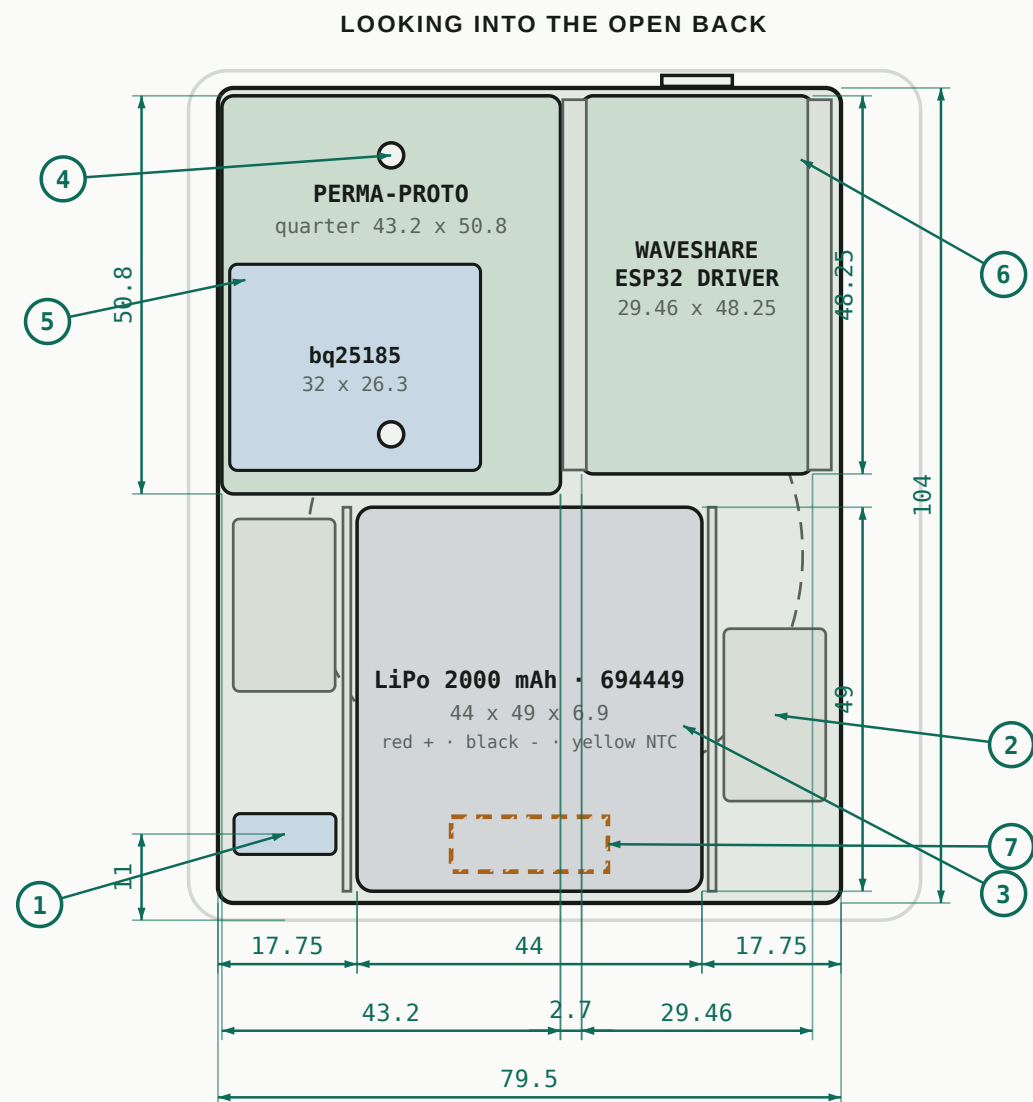
Every dimension on these sheets is generated from the OpenSCAD model's own parameters, so the drawings cannot drift from the exported STLs. Re-run `tools/mkdrawings.py` after any parameter change and the sheets follow.

Envelope **93.4 × 108.4 × 25** mm Ink area **63.6 × 84.8** mm Lean **20.0°** both orientations Filament **~104 g** Units **mm**

SHEET 1

Assembly & envelope Overall size, where the ink lands in the bezel, and every opening in the shell.





NOTES

- 1 USB-C charge breakout, on edge in printed rails, receptacle flush with the back cover. Two wires up the -X wall to the bq24074
- 2 module retention pads 13 x 22, one each side of the cell. Foam tape on the faces takes up the tolerance stack
- 3 stand puck, dia 63.3, bulges 5.95 into the interior behind everything on this sheet
- 4 2 x M2.5 posts, 35.6 apart, behind the Perma-Proto; the bq24074 rides on its front face
- 5 bq25185 charger breakout 32 x 26.3 x 7.2, low end of the Perma-Proto, nearest the charge port
- 6 driver-board rails 3 wide, overlapping the PCB 2 each side
- 7 MODULE 8-PIN HEADER KEEP-OUT 20 x 7 x 9 deep, position ASSUMED. Clashes with the cell by 1.1 - DO NOT PRINT, see below

MEASURE BEFORE YOU PRINT

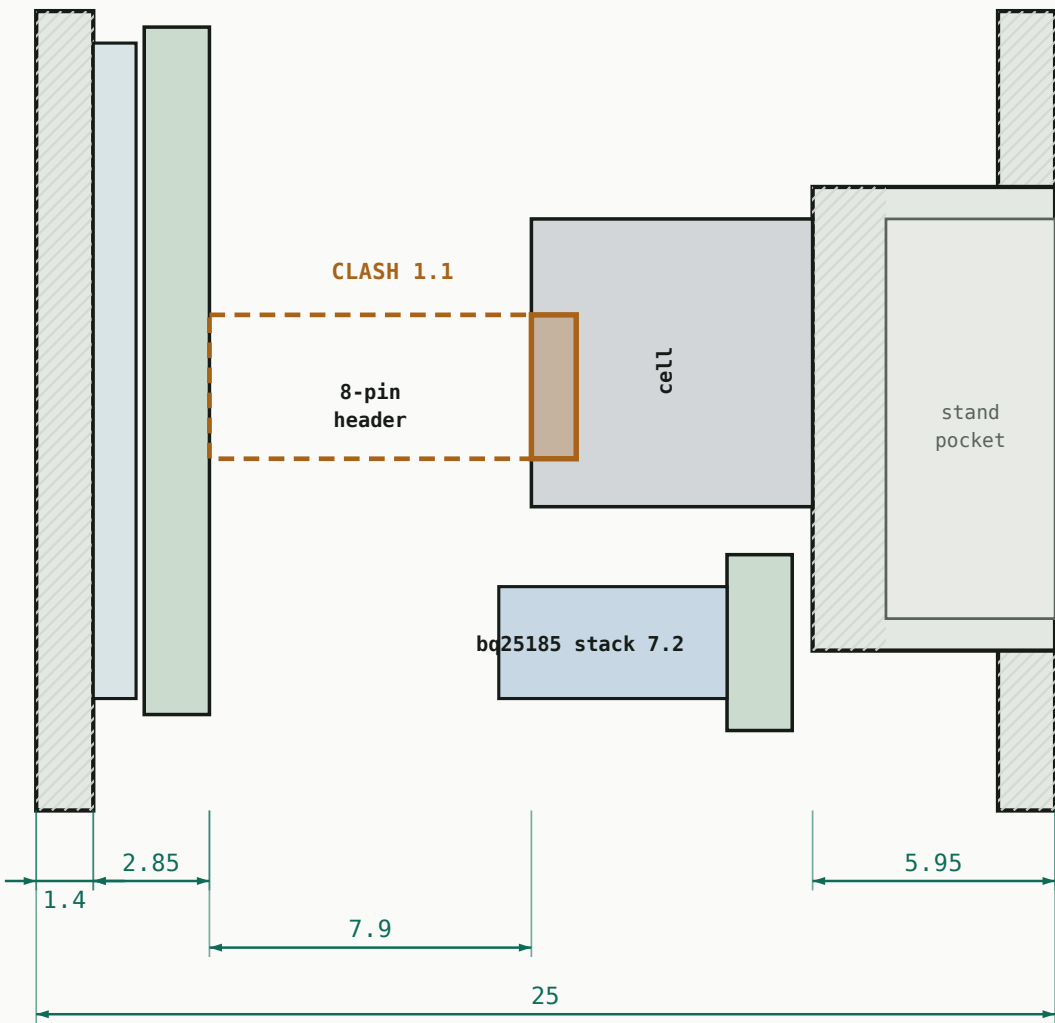
Three numbers on the module settle this sheet:

- a header edge, and where across X
conn_dz, conn_dx
- b header stand-off, mated, with the bend
conn_h (assumed 9)
- c is the glass centred on the PCB in Z?
pan_off_z, act_off_z

The board band clears a 9 mm header (12.7 available); the cell band does not (7.9). If the header is on the cell's edge the two bands swap, which also moves the flash port to the opposite wall. Or keep this layout and set depth = 26.6.
STLs export flat-face-down and MIRRORED IN X (print_mirror = true)

INTERNAL LAYOUT		
UNITS mm · SCALE 1:1	SHEET 2/8	REV E

SECTION A-A · depth axis magnified x5.2



DEPTH BUDGET

front face to module back	4.25
clear in front of the cell	7.9
clear in front of proto PCB	12.7
...less the bq25185 stack	7.1
clear in front of driver PCB	12.7
interior, outside the pocket	19.35
interior, over the pocket	14.8
total depth	25

THE TIGHT ONE

A mated vertical PH2.0 plug plus a wire bend is 8-9 mm. Over the boards there is 12.7 and it fits. Over the cell there is 7.9 and it does not - the keep-out overruns the cell face by 1.1.

Three ways out. Which is right depends on where the header actually is - measure it, see sheet 2:

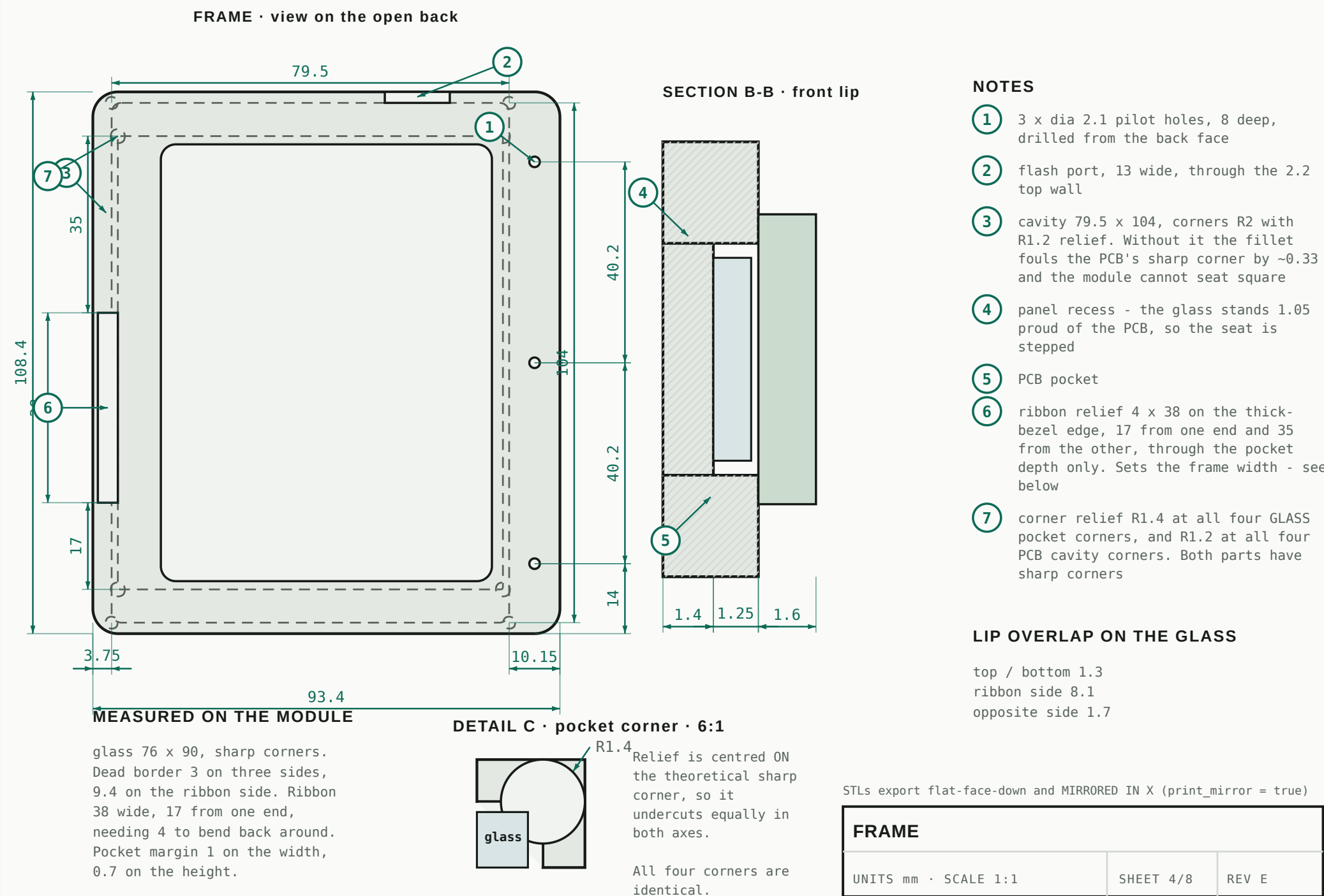
- 1 put the boards in whichever band the header lands in, cell in the other. No thickness cost, but it moves the flash port to the far wall and the charge port out of the board band.
- 2 right-angle PH2.0 housing, or desolder the header and lay the eight wires flat. Keeps depth 25; needs conn_h <= 7.4.
- 3 set depth = 26.6 and re-export. Everything downstream follows. On the PCBs (print_mirror = true) do not export the face down. On the PCBs (print_mirror = true)

DEPTH STACK-UP

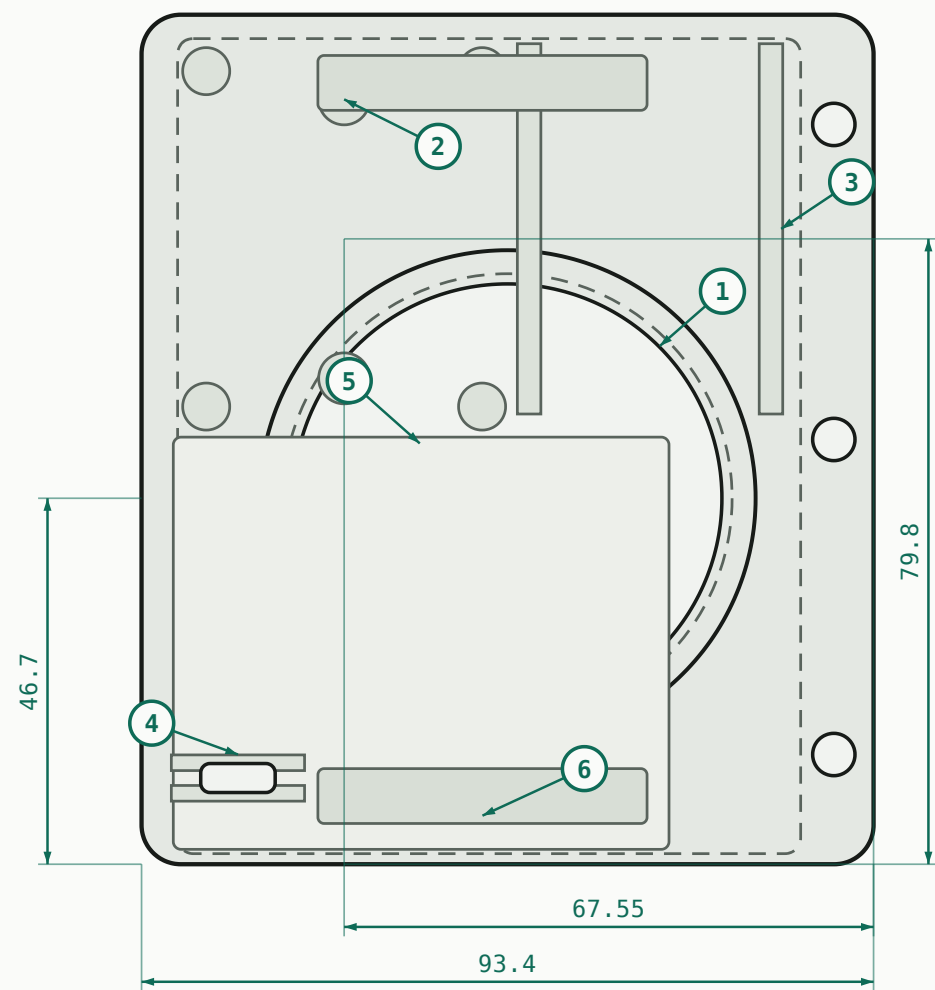
UNITS mm · SCALE SEE NOTE

SHEET 3/8

REV E



BACK COVER · inner face

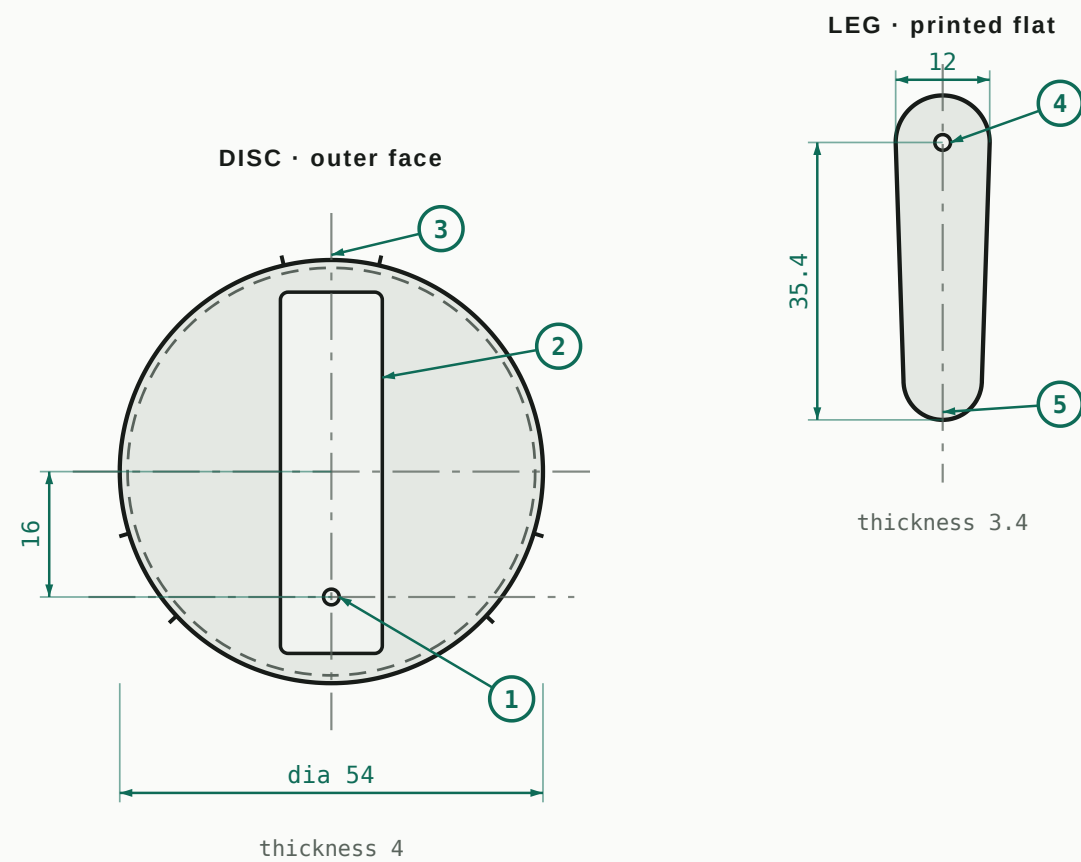


NOTES

- 1 pocket bore dia 54.7 x 4.15 deep. Bayonet groove dia 57.3 with three 34 deg entry gaps at 45 / 165 / 285 deg
- 2 2 x M2.5 posts, 35.6 apart, plus four corner pads
- 3 driver-board rails - the board slides down from the top, stop at the bottom
- 4 USB-C breakout rails; the port opening is through the skin below them
- 5 cell platform, level with the puck face so the cell does not straddle the step, with a 1 fence
- 6 module retention pad

STLs export flat-face-down and MIRRORED IN X (print_mirror = true)

BACK COVER		
UNITS mm · SCALE 1:1	SHEET 5/8	REV E



NOTES

- 1 dia 2.25 pin hole, straight through the disc. Pin is trapped by the pocket wall once the disc is fitted
- 2 leg slot 13 wide, from 23.2 below the centre to 22.9 above
- 3 three bayonet lugs, 26 deg wide x 1.3 proud, on the inner face
- 4 dia 2.15 axle hole. Use a 2 rod or a 1.75 filament offcut
- 5 foot R5. The heel behind the pin is cut by the pocket floor at the 58 deg stop, which is what takes the load

STLs export flat-face-down and MIRRORED IN X (print_mirror = true)

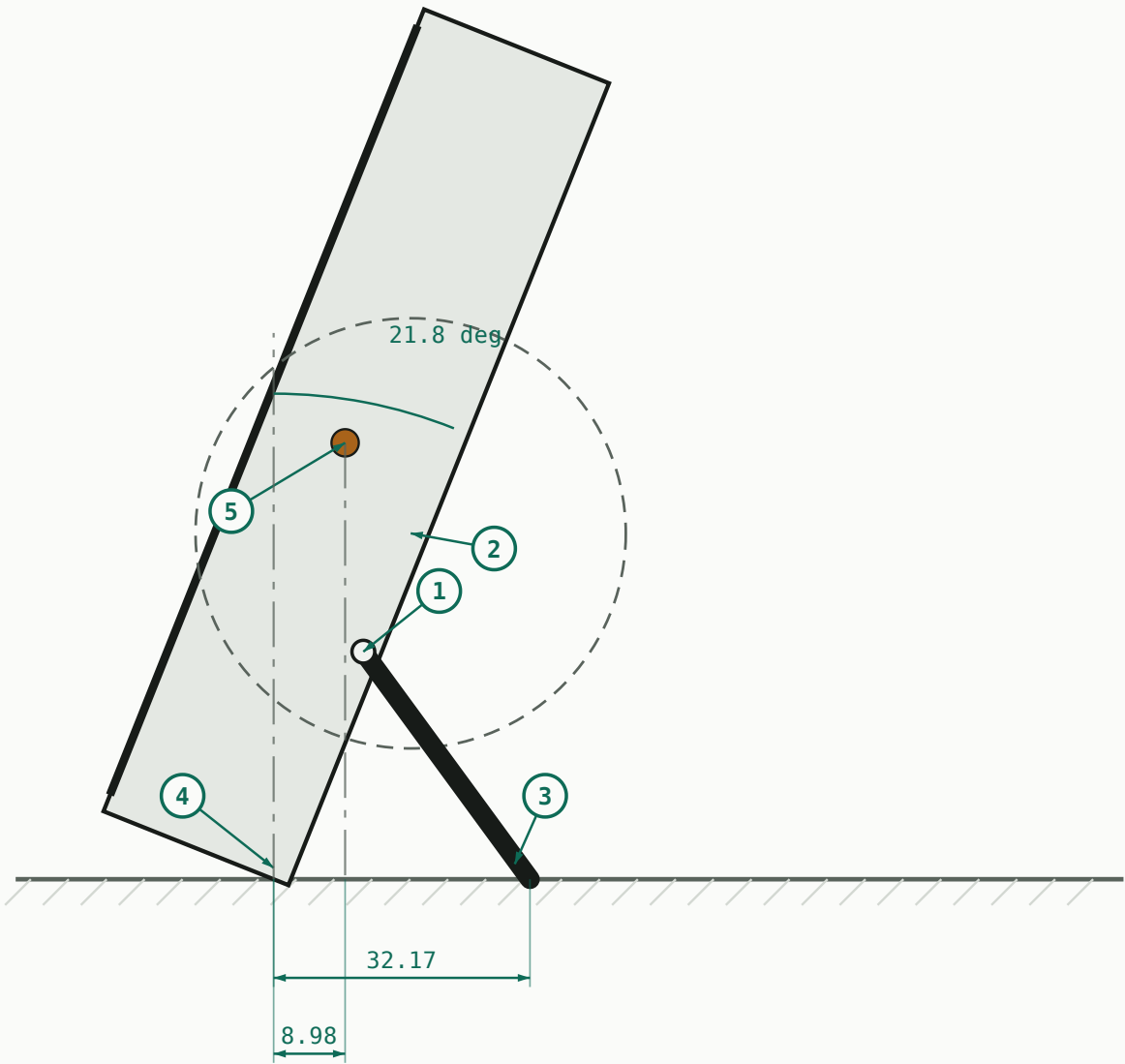
DISC & LEG

UNITS mm · SCALE 1:1

SHEET 6/8

REV E

SIDE ELEVATION · leg deployed



footprint / CoM offset, both behind the contact edge

NOTES

- 1 hinge pin, 16 from the hub centre
- 2 disc hub, 46.7 above the bottom edge
- 3 foot. Leg 35.4 long, hard stop at 58 deg from straight down
- 4 contact edge - the rear edge of the flat bottom, brought forward 2 by the back chamfer. This, not the front edge, is what the frame leans on
- 5 estimated centre of mass. It must sit behind the contact edge or the frame simply falls flat onto its face

WHY THE HUB SITS LOW

The frame is 93.4 x 108.4. A hub on the frame's centre would sit 54.2 above the bottom edge in portrait but only 46.7 in landscape, so the leg over-reaches and landscape stands about 9 deg more upright - close to tipping forward.

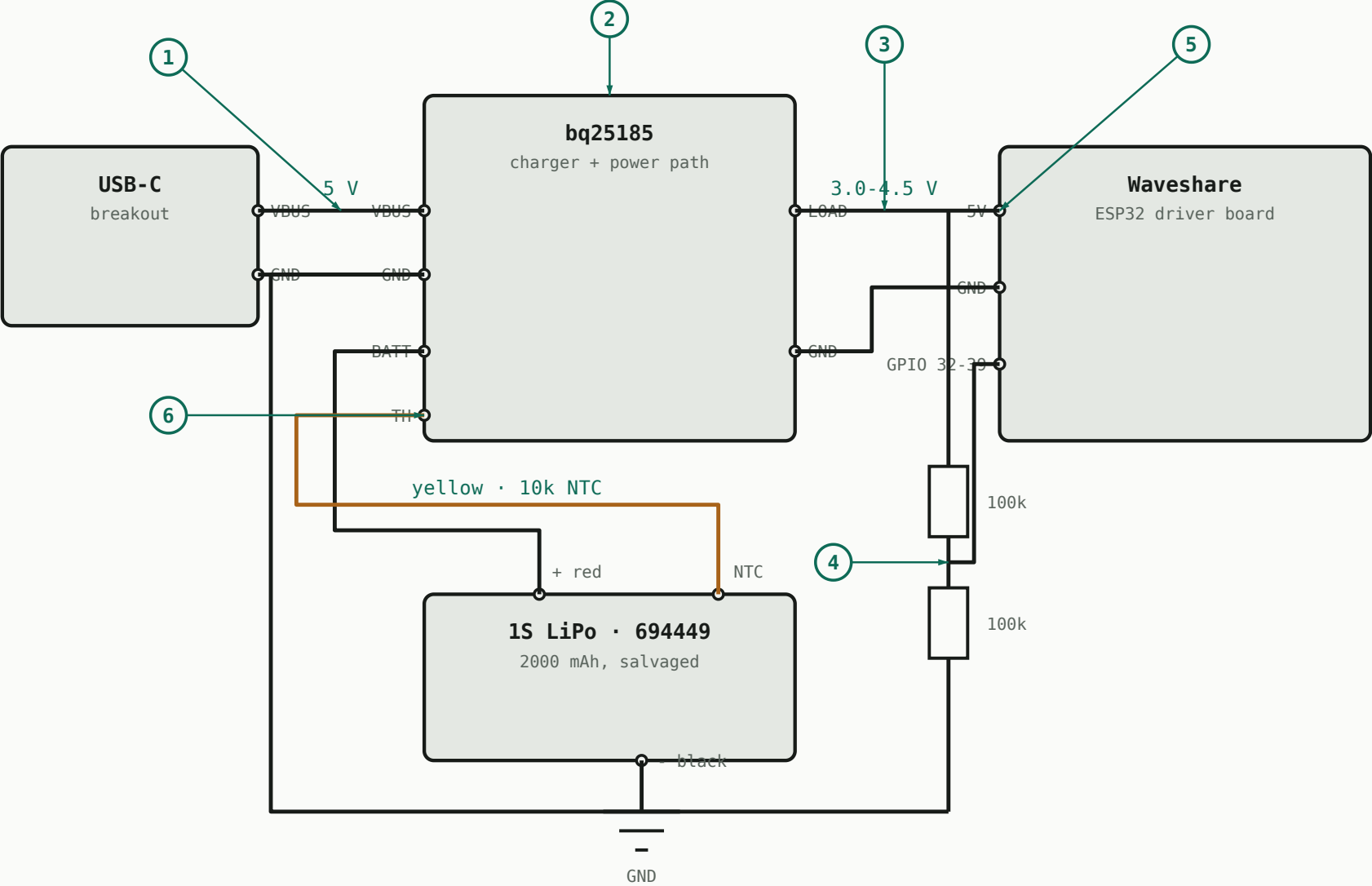
Putting the hub at $z = 46.7 = W/2$ makes the pivot the same height above whichever edge is down. Both orientations then give:

lean 21.8 deg · footprint 32.17

STLs export flat-face-down and MIRRORED IN X (print_mirror = true)

STAND KINEMATICS		
UNITS mm · SCALE 1:1.05	SHEET 7/8	REV E

POWER WIRING · charge and run from one cell



NOTES

- 1 USB-C breakout needs 5.1k from CC1 and CC2 to GND or a USB-C source will never turn its 5 V on. VBUS and GND only, no data
- 2 charge current: Adafruit's pinout page says it ships at 500 mA, the product page says 1 A - CHECK THE BOARD. 1 A is 0.5C for this cell and is what you want. Solar capability is input voltage regulation and is inert on USB
- 6 THREE WIRES = the pack has an NTC. Cut the TH jumper before wiring yellow here (THERM on the 4755). Verify first: red positive to black at 3.0-4.2 V, and yellow-to-black ~10k falling as you warm it. If it does not move it is not a thermistor: leave the jumper intact. Check for a protection PCB at the tab end; without one there is no short or overdischarge protection

- 3 LOAD is 4.5 V max - from the input when plugged in, from the cell otherwise. The 5V pin takes 3.6-5.5 V and browns out near 3.6 V
- 4 divider must land on ADC1 (GPIO 32-39) - ADC2 is dead while WiFi is on. Below 3.6 V firmware MUST deep-sleep for good, or the cell gets dragged to its protection cut
- 5 PWR / CHRГ / FAULT LEDs onboard; S1 and S2 pads mirror them. /CE pulled high disables charging

STLs export flat-face-down and MIRRORED IN X (print_mirror = true)

POWER WIRING

UNITS mm · SCALE N.T.S. SHEET 8/8 REV E

BEFORE YOU PRINT

Five numbers the model assumes rather than knows

Everything else on these sheets is either a published spec or falls out of the geometry. These five came from inference, so they are the ones to check with calipers.

act_off_x 3.2 mm	Offset of the ink centre from the PCB centre across the short axis. Inferred from panel vs active sizes — Waveshare publishes no datum. Wrong here and the white line goes lopsided. This is what <code>bezel_test.stl</code> checks.
conn_h 9 mm	Assumed height of the module's mated 8-pin header off its back. Sheet 3 shows 7.9 mm available over the cell. A stock vertical PH2.0 plug plus a wire bend is 8–9 mm — right-angle housing, direct-solder, or <code>depth = 25</code> .
pan_off_x / pan_off_z 0	Glass assumed centred on the PCB, leaving 6 mm of bare PCB at each end of the long axis. Worth a caliper check — it moves the panel recess on sheet 4.
proto_hole_sp 35.6 mm	Mounting-hole spacing on the quarter-size Perma-Proto (1.4"). The two M2.5 posts on sheet 5 sit on it.
USB-C positions centred	Assumed centred on the driver board's short edge and on the bq24074. Shifts the top-edge flash port if not.

Printed parts

PART	FILE	ON THE BED	VOLUME	FILAMENT
Frame	<code>frame.stl</code>	front face on the bed	38.7 cm ³	~48 g
Back cover	<code>cover.stl</code>	outer face on the bed	37.1 cm ³	~46 g
Disc	<code>disc.stl</code>	outer face on the bed	6.7 cm ³	~8 g
Leg	<code>leg.stl</code>	flat	1.3 cm ³	~2 g
Bezel test tile	<code>bezel_test.stl</code>	flat — print first	12.4 cm ³	~15 g

PLA or PETG, 0.2 mm layers, 4 perimeters, 15–20 % infill, no supports. The bayonet groove and the window chamfer are both cut at 45°, and the port openings bridge ≤13 mm.

Everything that goes inside

ITEM	SIZE / NOTE
Waveshare 4.2inch e-Paper Module (B)	78.5 × 103 × 1.6 PCB, panel 76 × 90 × 1.05
Waveshare e-Paper ESP32 Driver Board	29.46 × 48.25 – runs from the cell via its 5V pin, but has no charger on it

ITEM	SIZE / NOTE
Adafruit Perma-Proto, quarter-size	43.2 × 50.8, holes 35.6 apart
Adafruit bq24074 charger (#4755)	32 × 26.3
1S LiPo, 1200 mAh	44 × 49 × 6.9
USB-C breakout, 5.1 kΩ CC pulldowns	~13 × 13
M2.5 × 8 self-tapping screws	3 off – back cover
M2.5 × 6 screws	2 off – Perma-Proto
ø2 × 43 rod or 1.75 filament offcut	1 off – kickstand pin

Generated from `src/epaper_stand.scad` rev D · all ten pairwise interference checks in that file return empty · five parts manifold and watertight.